



AMIR SUHAIL SALIM

AI, Robotics & Control Engineer | M.Sc. Power Engineering, TUM

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Master's student at TUM building intelligent systems across AI, robotics, autonomy, and control. Experience spans learning-based control, NMPC, reinforcement/imitation learning, ROS 2 robotics, UAV trajectory tracking, FR3 manipulation, optimization, and tested engineering software. Strong at turning research models into reproducible pipelines and hardware-aware implementations.

CORE SKILLS

Programming: Python, C++, C, MATLAB, Pandas, NumPy

Robotics: ROS 2, MoveIt 2, RViz, Gazebo, MuJoCo, OpenCV, Intel RealSense, Franka FR3, UR10e

Optimization: CasADi, IPOPT, Pyomo, Gurobi, AC/DC OPF, MILP, genetic algorithms

AI / ML: PyTorch, imitation learning, behavior cloning, DAgger, SAC, XGBoost, LSTM

Control: MPC, NMPC, stochastic MPC, PID, LQR/LQ, H-infinity, SMC, observers, system identification

Engineering Tools: Simulink, Docker, Git, Pytest, CI/CD, CAN, MQTT, CoAP, HTTP

RESEARCH & ENGINEERING EXPERIENCE

Master's Thesis - Learning-Based End-Effector Trajectory Tracking for UVMS

Jun 2026 - Present

Munich Institute of Robotics and Machine Intelligence, Technical University of Munich

- Researching learning-based trajectory tracking for a BlueROV2 Heavy with a Reach Alpha 5 manipulator.
- Initializing neural control from a real-system-validated TPIK + PID-C expert via behavior cloning, with limited DAgger to reduce closed-loop distribution shift.
- Planning simulation-only SAC fine-tuning with focused domain randomization and a deterministic safety layer for staged real-system validation.

Research Intern - Imitation Learning for Optimal Thrust Control of Rocket Engines

Jan 2026 - Apr 2026

Chair of Automatic Control Engineering, Technical University of Munich

- Built a reactive NMPC expert in CasADi/IPOPT and generated 10,000 demonstrations from 125 closed-loop episodes across the 9-15.5 bar operating range.
- Trained a compact 56-parameter behavior-cloning policy with weighted MSE, reaching 0.002 held-out valve-command RMSE and NMPC-comparable pressure tracking.
- Reduced mean controller computation time from 60.6 ms to 0.32 ms, about 190x faster, and analyzed robustness to out-of-distribution initial conditions and pressure references.

Python Test Automation Project

Jan 2025 - Feb 2025

Munich Electrification, Germany

- Developed Pytest-based automated test frameworks, test specifications, and acceptance criteria for engineering software modules.
- Automated logging and report generation, improving review speed and reducing manual testing effort.

SELECTED PROJECTS

ATLAS CONOPS - RF Mesh Network Coverage Planner

Built a Streamlit coverage-planning tool at the European Defence Tech Hackathon 2026 for drone-detection network placement around Munich Airport.

Python, Streamlit, PSO, RF link budget, SRTM terrain LOS, Pydeck, GeoPandas, GIS export

Franka FR3 Task Board Solution

Developed a vision-driven FR3 system to detect and press illuminated buttons using HSV/template matching, RealSense depth, FSM logic, and MoveIt 2 planning.

ROS 2 Humble, MoveIt 2, Franka FR3, OpenCV, RealSense, Docker, Python

Autonomous Drone Racing

Implemented MPC-based UAV trajectory tracking with real-time obstacle avoidance and dynamically feasible trajectory generation.

ROS 2, Python, MPC, trajectory optimization, UAV control

ADDITIONAL PROJECTS

Advanced Control Systems Lab

Completed eight hardware experiments covering stochastic MPC, H-infinity mixed-sensitivity control, LQR, PID, SMC, Luenberger observers, and system identification.

MATLAB, Simulink, HIL, control design, model validation

Wind Power Prediction

Forecasted wind power output from multi-height meteorological data and analyzed model behavior across operating conditions.

Python, XGBoost, forecasting, Jupyter

Drawbot

Built a UR10e drawing pipeline from image vectorization through TCP calibration, RoboDK trajectory simulation, and real robot execution.

UR10e, RoboDK, Python, TCP calibration, trajectory planning

Energy Optimization Portfolio

Implemented unit commitment, AC/DC optimal power flow, and distribution-grid DER placement and sizing with optimization-based workflows.

Gurobi, Pyomo, NEOS, pandapower, MILP, AC/DC OPF, genetic algorithm

EDUCATION

Master of Science - Power Engineering

Oct 2023 - Present

Technical University of Munich, Garching

- Focus: control systems, autonomous robotics, model predictive control, system modelling, reinforcement learning, energy systems.
- Relevant coursework: Advanced Control, Advanced Control and Robotics Lab, Robot Learning and Control Project Course, Autonomous Vehicles, Energy Informatics, Battery Storage.

Bachelor of Engineering - Mechanical Engineering

Jun 2017 - Dec 2021

Vellore Institute of Technology, Vellore, India

- Thesis: Design Optimization and Numerical Analysis of Battery Thermal Management Systems for Electric Vehicles.
- Member of KI Racing, Baja SAE student team; supported front-wheel assembly design and manufacturing coordination.

EARLIER EXPERIENCE & LANGUAGES

Student Intern: Hindustan Machine Tools, Kochi, India - CNC machining, G-code programming, manufacturing systems.

Languages: English fluent professional; German basic (A2, improving); Malayalam native.